Day-2

Power supply Systems Theory

- Basics of SMPS and UPS sytems.
- Identify power cables and connectors.

Q 1 :- List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Ans - **SMPS and UPS – Computer Power Supply**

1. Three devices that use SMPS and two devices that use UPS

SMPS (Switched Mode Power Supply): It converts electricity into the correct voltage needed by electronic devices.

UPS (Uninterruptible Power Supply): It provides temporary backup power when the main electricity supply goes off.

A. Three devices that use SMPS

Device	**Why it uses SMPS**
Desktop computer	It converts electricity into different voltages needed by computer parts.
Laptop charger	It converts high-voltage electricity into low-voltage power for charging the laptop.
LED TV	It provides the correct power needed by the TV's electronic circuits.

B. Two devices that use UPS

Device	**Why it uses UPS**
Desktop computer	It provides backup power and helps prevent sudden shutdowns.
Wi-Fi router	It keeps the internet working for some time during a power cut

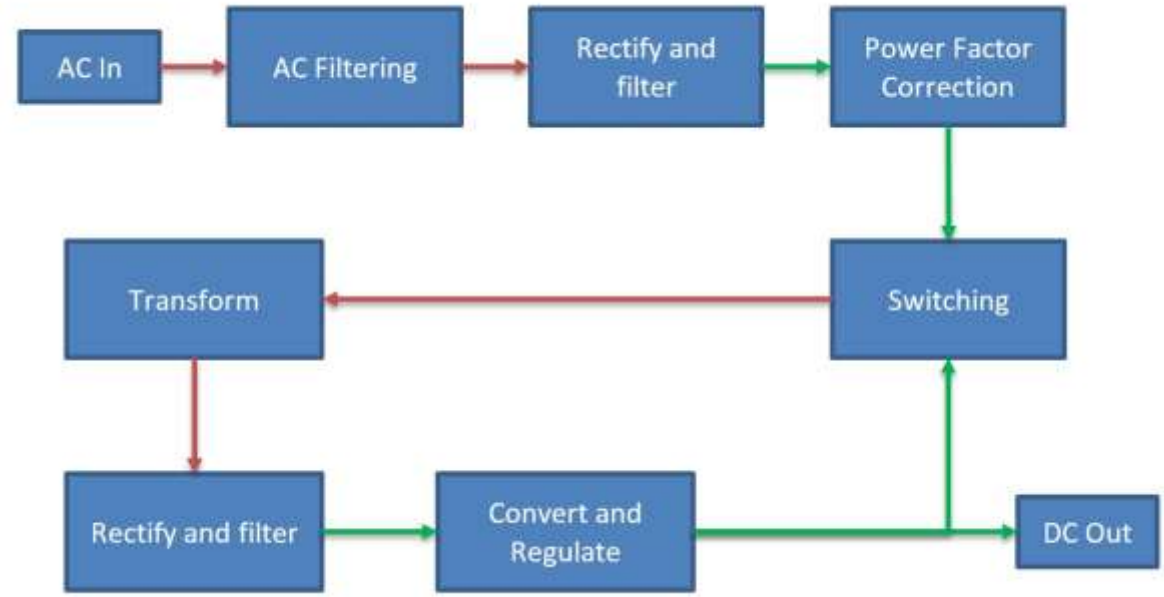
Q 2 :- Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

Ans- **2. Diagram: Internal components of an SMPS**

An SMPS is a power supply unit (PSU) found inside a desktop computer. It converts AC electricity into DC electricity.

Labeled SMPS block diagram



1. AC Input

Electricity from wall socket.

2. Rectifier

Converts AC into DC.

3. Filter and Switching Circuit

Smooths and switches the electrical power.

4. High-Frequency Transformer

Changes voltage and provides isolation.

5. Output Rectifier and Filter

Produces smooth DC electricity.

6. DC Output

3.3V, 5V and 12V for computer parts.

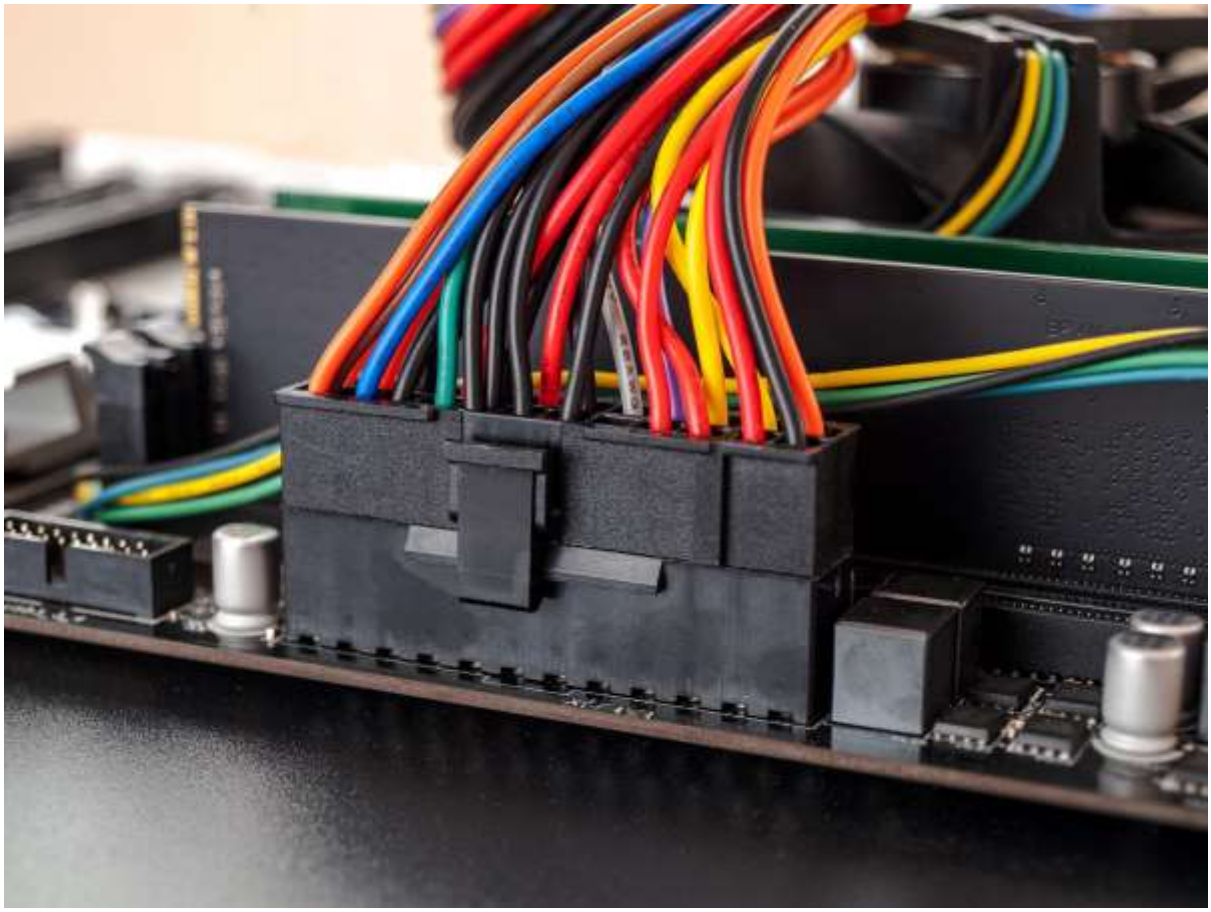
Explanation of components:

1. AC Input – Receives electricity from the wall socket.
2. Rectifier – Changes AC electricity into DC electricity.
3. Transformer – Changes the voltage to the required level.
4. Output Rectifier – Converts the transformed power into DC.
5. Filter – Makes the output power smooth.
6. DC Output – Supplies power to the motherboard, graphics card and other parts.

Q -3 :- Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Ans- **Four types of SMPS power connectors.**

(1) - ATX 24-pin motherboard connector



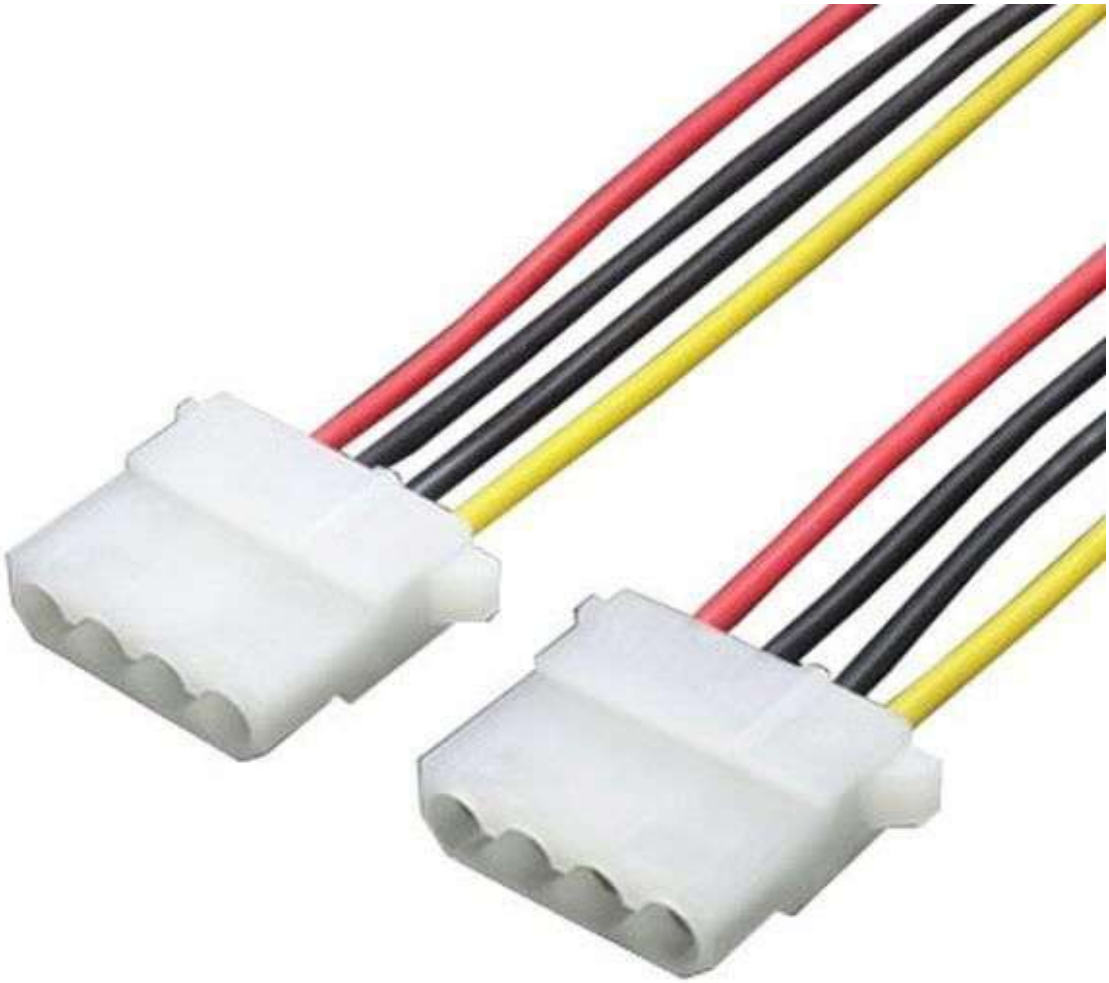
- **Name: ATX 24-pin connector.**
- **Used for: Supplies main power to the motherboard.**
- **Safety tip: Check the correct direction and push gently until the clip locks.**

(2) - SATA power connector.



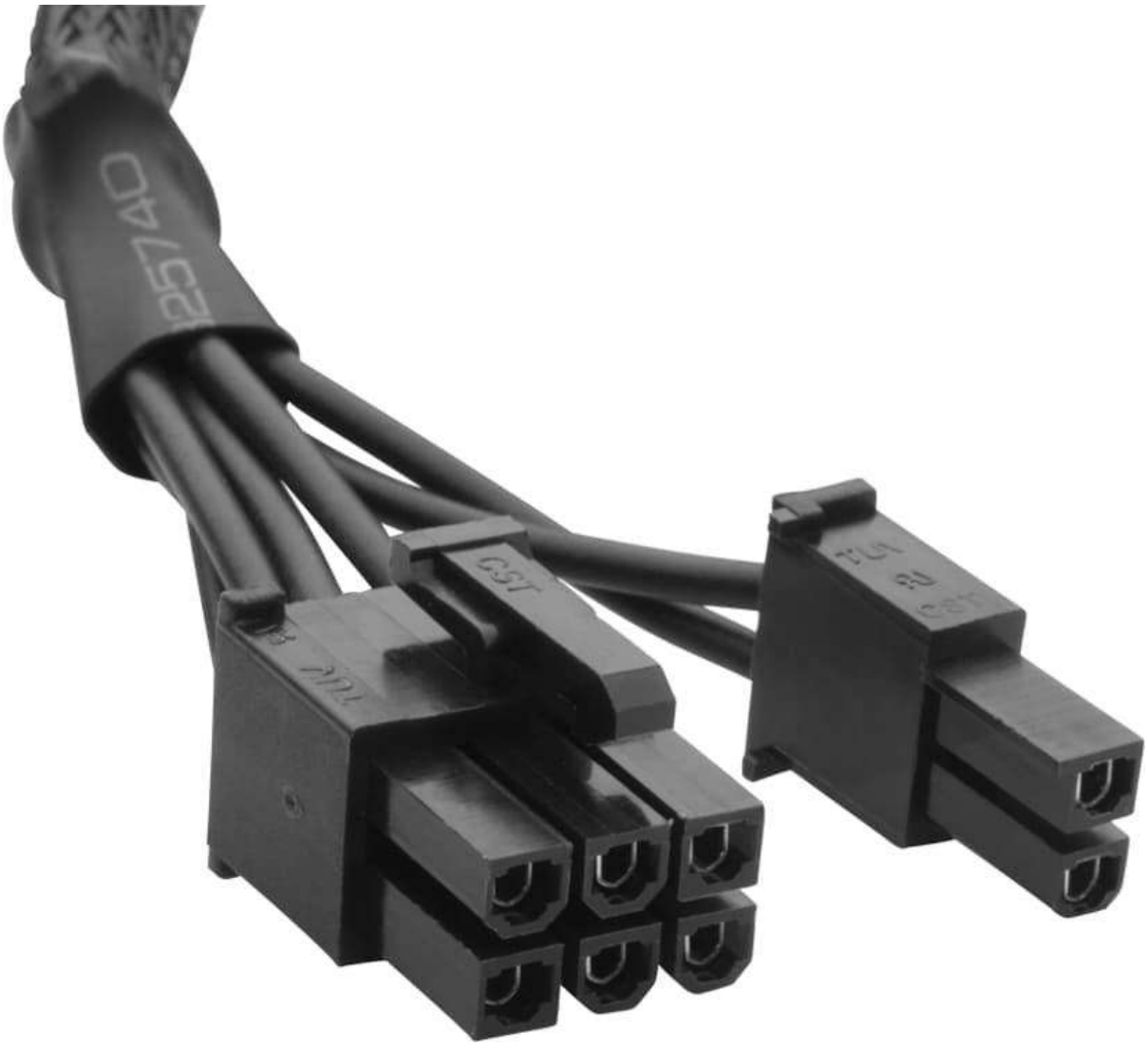
- **Name:** SATA power connector.
- **Used for:** Supplies power to SSDs and hard disk drives (HDDs).
- **Safety tip:** Match the L-shaped connector before inserting it. Never force it.

3 - **Molex connector**



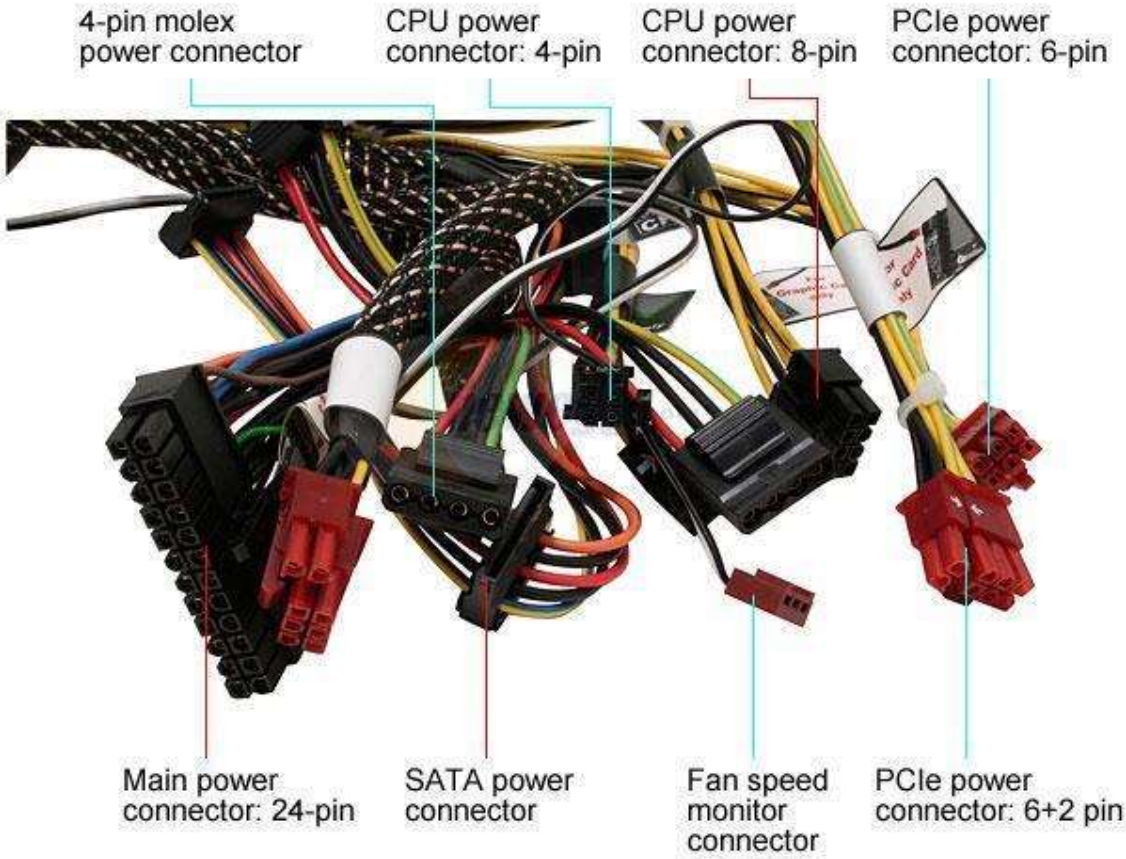
- **Name:** 4-pin Molex connector.
- **Used for:** Powers older hard drives, fans and some accessories.
- **Safety tip:** Hold the plastic connector while removing it. Do not pull the wires.

4 - PCIe 6/8-pin connector



- **Name:** PCIe 6-pin or 8-pin connector.
- **Used for:** Supplies extra power to a graphics card (GPU).
- **Safety tip:** Use the correct PCIe cable and make sure the connector is fully locked.

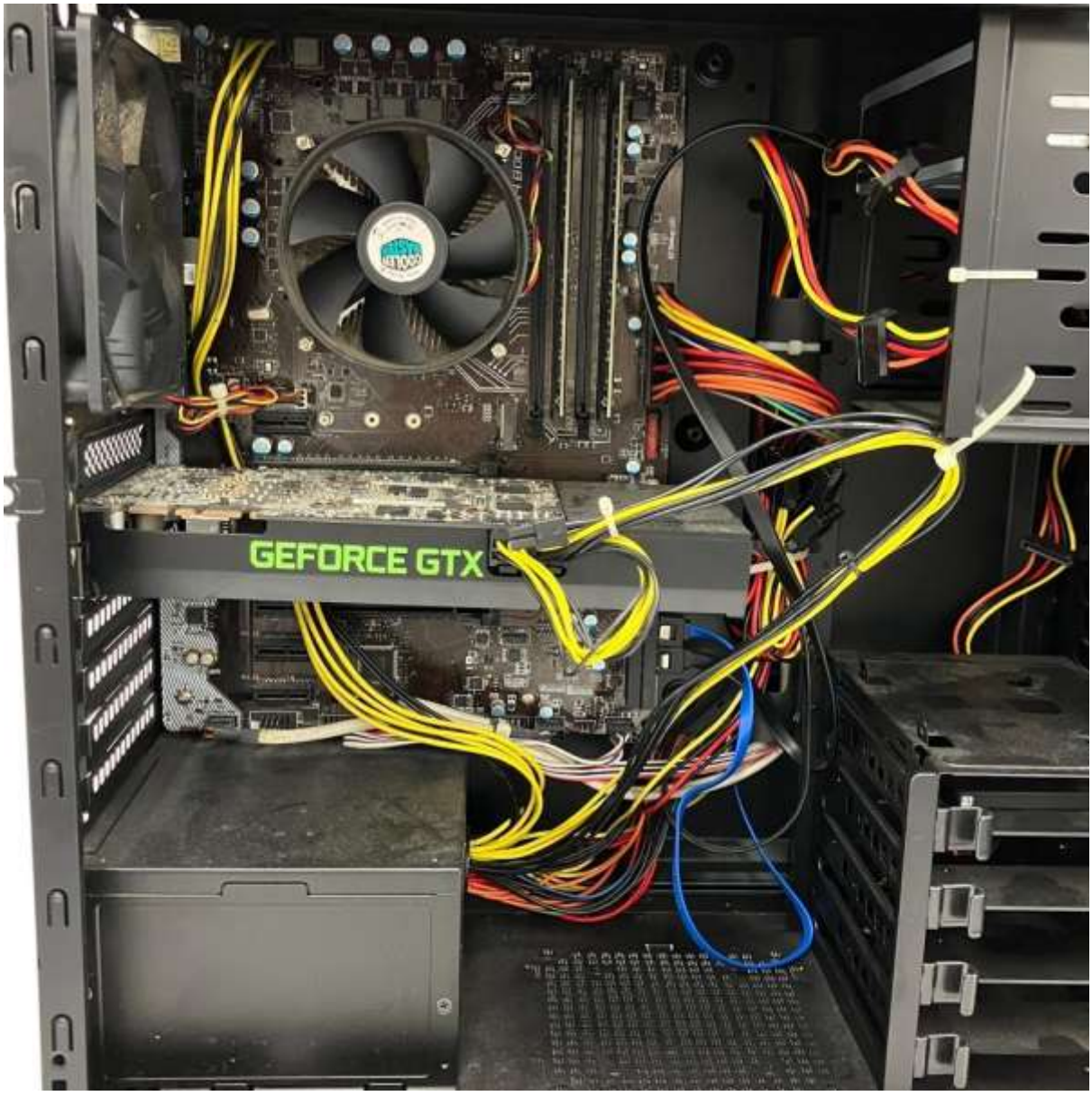
All types of coonectors are as under.



Q -4 :- Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Ans-**Steps to connect SMPS in a new gaming PC**

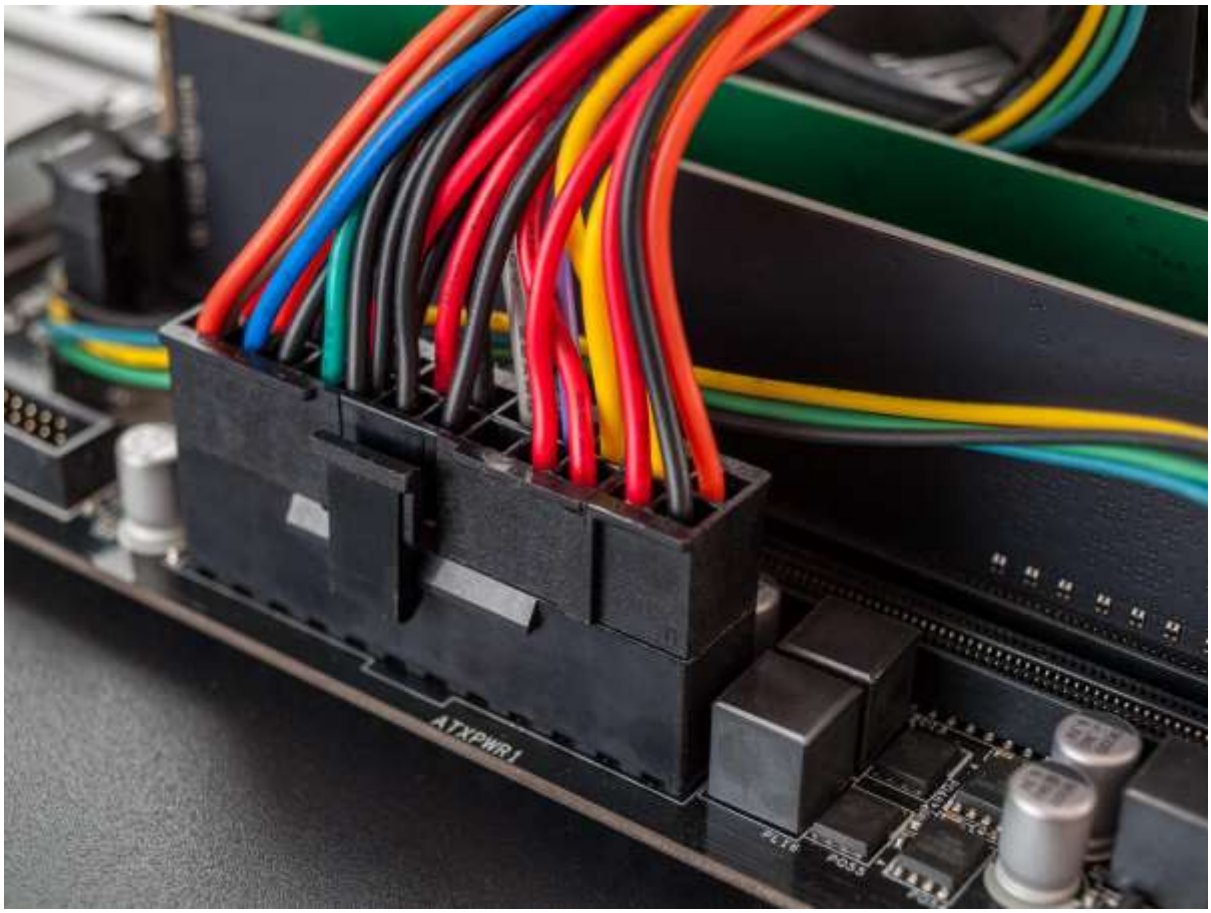
Suppose I am building a new gaming computer. I will follow these steps to connect the SMPS to the motherboard, graphics card and storage devices.



Step 1: Install the SMPS

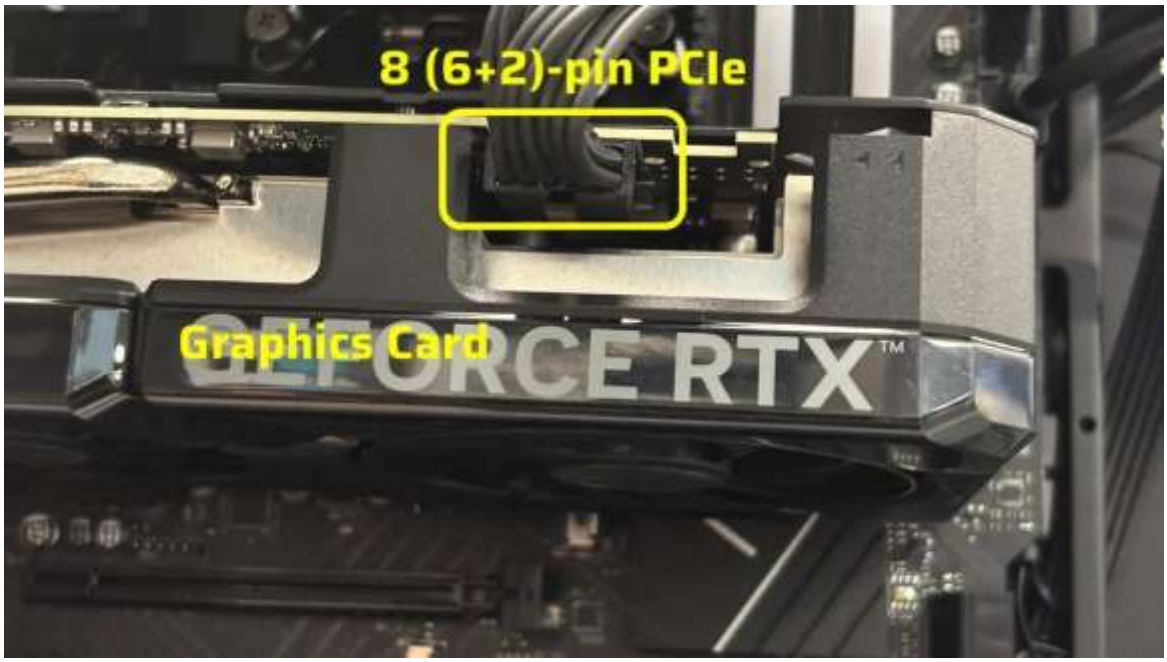
- Place the SMPS inside the computer cabinet.
- Fix it using screws.
- Keep the fan facing the correct ventilation opening.

Step 2: Connect the motherboard.



- Find the 24-pin ATX cable from the SMPS.
- Connect it to the 24-pin motherboard socket.
- Connect the 4+4-pin CPU/EPS cable to the CPU power socket near the processor.

Step 3: Connect the graphics card



- Find the PCIe 6+2-pin or 8-pin power cable.
- Connect it to the graphics card.
- Some modern graphics cards use a 12V-2x6 connector instead, so follow the graphics card manual.

Step 4: Connect the storage devices.



- Find the SATA power cable from the SMPS.
- Connect it to the SATA SSD or HDD.
- Connect a separate SATA data cable from the drive to the motherboard.

Step 5: Check all connections

- Check that all connectors are properly fitted.
- Make sure no cables are touching the fans.
- Close the computer cabinet.

Step 6: Turn on the computer

- Connect the power cable to the wall socket.
- Switch on the SMPS.
- Press the computer's power button.
- Check that the computer starts properly.

Result The SMPS supplies power to the motherboard, graphics card and storage devices, and the gaming PC is ready to use.

Important safety precautions

- Always switch off and unplug the computer before connecting or disconnecting power cables.
- Never open the SMPS metal box because it can contain dangerous stored electrical charge.
- Use only the modular cables supplied for your exact PSU model. Modular PSU cables are not always interchangeable.

Conclusion -

SMPS provides the required power to computer components, while UPS provides backup electricity during power cuts. Different connectors such as ATX, SATA, Molex and PCIe help supply power to different computer parts.